Write a C program to check whether a given string belongs to the language defined

by a Context Free Grammar (CFG)

S → 0S1 | ε

AIM: To write a C program to check whether a string belongs to the language defined by the CFG

S -> 0S1 | e .

CODE:

Programiz
C Online Compiler

main.c

Run ▶

```c
#include <stdio.h>
#include <string.h>
int main() {
    char s[100];
    int i, n, flag = 1;
    printf("Enter a string: ");
    scanf("%s", s);
    n = strlen(s);
    if (n % 2 != 0)
        flag = 0;
    else {
        for (i = 0; i < n / 2; i++) {
            if (s[i] != '0' || s[n - 1 - i] != '1') {
                flag = 0;
                break;
            }
        }
    }
    if (flag)
        printf("String belongs to the language");
    else
        printf("String does not belong to the language");
```

OUTPUT:

Output

```
Enter a string: 0101
String does not belong to the language
```

RESULT: The C program checks whether the given string belongs to the language defined by the CFG.